

llmXive Automated Review of Imaginative Perception Tokens Enhance Spatial Reasoning in Multimodal Language Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper’s central claims regarding state-of-the-art performance and the efficacy of Imaginative Perception Tokens (IPT) are fundamentally undermined by the citation of non-existent or future-dated models as baselines and training sources.

Specifically, the results tables (Table 1 in the main text, Table `tab:pt_breakdown` in the supplement) list “GPT-5”, “GPT-5.1”, “Gemini 2.5 Flash”, “Gemini 3 Flash”, “Qwen3-VL-8B”, and “InternVL3.5-8B” as comparison points. As of the current date, none of these models exist in the public record. GPT-5 has not been released, Gemini’s latest major version is 1.5, and Qwen/InternVL are currently at version 2.x. The inclusion of these models renders the reported accuracy percentages (e.g., GPT-5 scoring 61.1% on EgoDir, Gemini 3 Flash scoring 96.2% on Real+Arr) unverifiable and scientifically invalid. A claim of “state-of-the-art” performance cannot be supported by comparisons against hallucinated baselines.

Furthermore, the methodology section states that “Text CoT generation” targets were generated by “GPT-5.1”. Since this model does not exist, the provenance of the training data is unclear. If the authors used a different model (e.g., GPT-4o), this must be corrected immediately. If the results for these baselines were simulated or hallucinated, the paper’s core contribution is invalidated.

While the internal consistency of the authors’ own model (BAGEL) across different training regimes appears mathematically consistent within the provided tables, the external validity of the paper’s conclusions is destroyed by the reliance on non-existent external references. The paper cannot be accepted until all baselines are replaced with real, verifiable models and the training data generation process is clarified with existing technology.

Action items.

- **[fatal]** The paper’s central claims regarding state-of-the-art performance and the efficacy of Imaginative Perception Tokens (IPT) are fundamentally undermined by the citation of non-existent or future-dated models as baselines and training sources. Specifically, the results tables (Table 1 in the main text, Table `tab:pt_breakdown` in the supplement) list “GPT-5”, “GPT-5.1”, “Gemini 2.5 Flash”, “Gemini 3 Flash”, “Qwen3-VL-8B”, and “InternVL3.5-8B” as comparison points. As of the current date, none of thes

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively provides a high-level overview of the three spatial imagination tasks (Path Tracing, Perspective Taking, Multiview Counting) as described in the caption. The layout clearly distinguishes between training examples (left) and evaluation examples (right) using distinct visual panels and headers, making the figure’s purpose immediately clear.

Figure 2

Figure 2 effectively displays three text CoT training examples for path tracing, clearly showing the input images, questions, and generated reasoning traces. The layout is uncluttered, and the content aligns perfectly with the caption’s description of the structured prompt and reasoning steps.

Figure 3

The figure effectively demonstrates the model’s reasoning process and success cases, but the third row’s generated image contradicts the caption’s claim of a ‘sideview’ and is spatially inconsistent with the provided inputs.

Figure 4

Figure 4 effectively displays dataset examples for the Path tracing task across real-world and synthetic environments. The layout is clear, with distinct sections for different settings (PathArr, Path, EgoDir) and consistent highlighting of correct answers, fully supporting the caption’s description.

Figure 5

Figure 5 effectively visualizes the perspective taking task with clear input, generated thought, and ground-truth novel viewpoint columns. The layout is uncluttered, and the use of green/red highlights for correct/incorrect predictions aligns perfectly with the caption’s description.

Figure 6

The figure displays perspective taking examples but fails to show all four sub-categories claimed in the caption, and the specific task labels are too small to read clearly.

Figure 7

Figure 7 effectively visualizes the multiview counting task by displaying input views, generated top-down maps, and ground-truth maps side-by-side. The inclusion of explicit question-answer

blocks with color-coded correctness indicators (green/red) aligns perfectly with the caption’s description. The layout is clear, and the visual comparison between the model’s imagination and the ground truth is legible.

Figure 8

Figure 8 effectively displays multiview counting dataset examples from AI2-THOR/ProcTHOR, clearly separating rotation and multi-camera trajectories. The layout is uncluttered, with input views and ground-truth maps clearly labeled, and the correct answers are highlighted in green as described in the caption.

Figure 9

The provided image for Figure 9 is a 3D scene rendering that completely contradicts the caption’s description of a VQGAN reconstruction quality analysis, making the figure unreadable and unusable for its intended purpose.

Figure 10

The figure is a rendering of a 3D scene that fails to match its caption, which describes a comparison between ground-truth and decoded tokens. The image lacks the necessary side-by-side layout, labels, and visual evidence to support the scientific claims made in the text.

Figure 11

The figure content is completely unrelated to the caption, displaying a 3D bathroom scene instead of the described grayscale decoded imaginative perception tokens.

Figure 12

The figure effectively demonstrates the impact of resolution on image quality, but the column headers appear to be ordered in reverse of the visual evidence and the caption’s description, creating significant confusion regarding which column represents which resolution.

Action items.

- **[writing]** Figure 3: The ‘Generated Thought’ column displays a top-down view in the third row, contradicting the caption’s description of generating a ‘visual thought (imagined sideview at M1)’.
- **[writing]** Figure 3: The third row’s ‘Generated Thought’ image depicts a living room with a sofa and rug, which is spatially inconsistent with the input top-down map and views showing a kitchen/dining area.
- **[science]** Figure 6: The caption claims to show examples from ‘four sub-categories’, but the figure only displays two categories (Distance and Position). The other two sub-categories mentioned in the caption are missing.

- **[writing]** Figure 6: The text labels for the sub-categories (e.g., ‘closer’, ‘further’, ‘left’, ‘right’) are extremely small and illegible in the rendered image, making it difficult to distinguish the specific task type for each row.
- **[fatal]** Figure 9: The rendered image is a 3D scene (room with table, chair, door) that does not match the caption’s description of a ‘VQGAN reconstruction quality’ analysis; the figure appears to be a rendering artifact or the wrong file.
- **[science]** Figure 10: The rendered image displays a single, unlabelled 3D scene (a room with a window and shadow) that does not visually correspond to the caption’s description of a ‘Ground-truth vs. decoded IPTs’ comparison. There are no side-by-side panels, no ground-truth reference, and no decoded output visible to support the claim of ‘visually degraded images’ or ‘limitations of discrete token generation’.
- **[writing]** Figure 10: The image lacks all necessary scientific annotation, including axis labels, a legend distinguishing ‘Ground-truth’ from ‘Decoded IPTs’, and any visual indicators (such as arrows or grid lines) to facilitate the comparison described in the caption.
- **[fatal]** Figure 11: The rendered image is a 3D bathroom scene (sink, wall, painting) that does not match the caption’s description of ‘grayscale representations’ or ‘decoded IPTs’ from a VLM; the figure content appears to be a rendering artifact or incorrect file.
- **[writing]** Figure 12: The column headers ‘Latent 64’, ‘Latent 32’, ‘Latent 16’, and ‘Latent 4’ contradict the caption’s description which states resolution increases from left to right (Latent-4 to Latent-64). The visual evidence shows the rightmost column is the blurriest (lowest quality), implying it corresponds to the lowest resolution (Latent-4), meaning the headers are either reversed or mislabeled relative to the visual data.
- **[writing]** Figure 12: The column headers do not match the caption’s terminology. The caption uses ‘Latent-4’ and ‘Latent-64’ (with hyphens), while the figure uses ‘Latent 4’ and ‘Latent 64’ (with spaces). While minor, consistency is preferred.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper generally maintains a high level of technical precision, but several acronyms and symbols specific to the authors’ experimental setup are introduced without definition, creating barriers for a competent reader from an adjacent field (e.g., NLP or general computer vision).

First, the abbreviation **TIFA** is used repeatedly in the Data Curation sections (e.g., “TIFA filtering”) without ever being expanded. While the context suggests it is a filtering mechanism, a reader outside the specific subfield of visual grounding benchmarks would not know if this refers to a specific metric, a tool, or a custom protocol. The acronym should be expanded at first use

(e.g., “Text-Image Faithfulness Assessment (TIFA)”) with a brief clause explaining its function.

Second, in the “Imaginative Token Exploration” section, the authors use **CB** and **f** as shorthand for “codebook size” and “downsampling factor” respectively (e.g., “CB 1K, f=16”). These are standard in VQ-VAE literature but are not defined in the text or table captions. A reader from an NLP background might not immediately map “CB” to codebook or “f” to the downsampling factor. These should be explicitly defined in the text or table footnotes.

Third, the term **HM3D** is used to describe the scene source for the Habitat experiments. While “Matterport3D” is mentioned earlier, the specific abbreviation “HM3D” (likely Habitat-Matterport3D) is not defined. Given that the paper distinguishes between “Matterport3D” (real-world) and “AI2-THOR” (synthetic), clarifying that HM3D refers to the specific subset or version of Matterport3D used in Habitat would prevent confusion.

Finally, **BEV** (Bird’s-eye view) is used in the Multiview Counting section. While common in autonomous driving, it is not a universal term in all multimodal reasoning papers. A one-time expansion (e.g., “Bird’s-eye view (BEV) map”) would ensure clarity for readers whose primary expertise is in language or general vision rather than robotics.

These are minor, easily fixable omissions that do not detract from the scientific contribution but are necessary for the paper to be self-contained for the target audience.

Action items.

- **[writing]** Section ‘Imaginative Token Exploration with Different VLMs’ (Supp. e002) uses ‘CB’ and ‘f’ in table headers and text (e.g., ‘CB 1K, f=16’) without defining them. Define ‘CB’ as ‘codebook size’ and ‘f’ as ‘downsampling factor’ at first use.
- **[writing]** Section ‘Data Curation Details’ (Supp. e001) references ‘TIFA filtering’ multiple times without defining the acronym. Expand to ‘Text-Image Faithfulness Assessment (TIFA)’ at first occurrence and briefly state its purpose (verifying object visibility/position).
- **[writing]** Section ‘Perspective Taking’ (Supp. e001) uses ‘HM3D’ as a shorthand for the dataset source without expansion. Define as ‘Habitat-Matterport3D (HM3D)’ at first use to clarify the relationship to the previously mentioned Matterport3D dataset.
- **[writing]** Section ‘Multiview Counting’ (Supp. e001) uses ‘BEV’ (Bird’s-eye view) without expansion. While common in robotics, it is not universal in general VLM literature; define at first use (e.g., ‘Bird’s-eye view (BEV) map’).

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, but there are two specific instances where the conclusions drawn in the text do not strictly follow from the data presented in the tables, or where numerical reporting creates ambiguity.

First, in Section “Imaginative Token Exploration with Different VLMs” (e001/e002), the authors state: “IPT consistently outperformed answer-only and Text CoT on Path Tracing.” This conclusion is supported by Table `tab:qwen_discrete` only partially. For the Qwen2.5-VL 3B model, the “Answer-only” baseline achieves 48.6% on Path Tracing (PT), while the best IPT configuration achieves 55.0%. This supports the claim. However, for the 7B model, the table lists “Answer-only” as 37.6% but does not provide a PT score for it (the cell is empty or the row is misaligned in the provided text representation, but the text implies a comparison). More critically, the text claims “consistent” outperformance, yet the table shows the 3B “Answer-only” score (48.6) is lower than IPT (55.0), but the 7B “Answer-only” score (37.6) is significantly lower than IPT (55.9). The logic holds *if* the 7B Answer-only PT score is indeed 37.6, but the table’s presentation of “–” for PET and the specific alignment of rows for 7B makes the “consistent” claim slightly fragile without explicit confirmation that the 7B Answer-only PT score is the 37.6 listed. A more precise issue is the comparison to Text CoT: the table shows Text CoT at 43.0% (3B) and 35.7% (7B), which IPT beats. The primary logical gap is the phrasing “consistently outperformed” when the 3B Answer-only baseline (48.6) is closer to IPT (55.0) than the 7B baseline (37.6) is to its IPT (55.9), and the table’s layout for 7B Answer-only PT is ambiguous in the source text provided. The claim should be qualified to reflect the specific baselines used or the table should be clarified to ensure the “consistent” narrative is mathematically supported by every cell.

Second, in the “Data Curation Details” for Path Tracing (AI2-THOR), the text states: “Training set includes ProcTHOR-10k.” Immediately following, the “Statistics” subsection reports: “11,204 synthetic examples.” The logical connection between “ProcTHOR-10k” (implying 10,000 scenes or examples) and “11,204 examples” is not explicitly defined. Does the 11,204 figure include the 10k plus additional samples? Or is “ProcTHOR-10k” a misnomer for a subset of the 11,204? This creates a minor inconsistency in the definition of the dataset size, which could confuse a reader trying to replicate the data volume. The text should clarify whether the 11,204 examples are derived from the 10k scenes or if the 10k is a subset of the 11,204.

These issues are primarily matters of precise reporting and logical entailment from the provided tables to the text claims, rather than fundamental flaws in the research hypothesis.

Action items.

- **[writing]** Section ‘Imaginative Token Exploration with Different VLMs’ claims ‘IPT consistently outperformed answer-only and Text CoT on Path Tracing’ based on Table ‘tab:qwen_discrete’. However, the table shows IPT (55.0/55.9) underperforming ‘Answer-only’ (48.6/37.6) for the 3B model in PT, and the 7B model has no ‘Answer-only’ PT baseline listed for comparison. The text’s claim of ‘consistent’ superiority is not entailed by the provided data table.
- **[writing]** In ‘Data Curation Details’ (Path Tracing, AI2-THOR), the text states ‘Training set includes ProcTHOR-10k’ but the ‘Statistics’ subsection immediately below reports ‘11,204 synthetic examples’. The relationship between the 10k ProcTHOR scenes and the 11,204 examples is ambiguous; it is unclear if the 11k count includes the 10k or if they are distinct subsets, creating a potential inconsistency in dataset composition reporting.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a compelling method for improving spatial reasoning via Imaginative Perception Tokens (IPT), but the rhetoric in the title, abstract, and conclusion frequently exceeds the scope of the demonstrated evidence.

First, the title “Imaginative Perception Tokens Enhance Spatial Reasoning in Multimodal Language Models” and the abstract’s claim of a general enhancement suggest a broad capability applicable to any spatial task. However, the empirical validation is strictly confined to three specific tasks (Path Tracing, Perspective Taking, Multiview Counting) and a narrow set of environments (primarily AI2-THOR, Habitat, and a specific subset of Matterport3D/VST). There is no evidence presented for outdoor navigation, complex 3D reconstruction, or general VQA tasks where spatial reasoning is not the primary focus. The claim should be scoped to the specific domains and tasks tested.

Second, the conclusion asserts that the method “enables models to internalize spatial reasoning.” This causal claim is not fully licensed by the results. The paper’s own visualization analysis (Section 5.1, Figure 3 caption) notes that the generated intermediate images “exhibit spatial imprecision” and “lack spatial structure,” yet the model still answers correctly. This discrepancy suggests the model may be learning to exploit dataset-specific shortcuts or correlations rather than developing a robust, internalized spatial model. The narrative should be adjusted to reflect that the method improves performance on these benchmarks, while acknowledging that the mechanism of “internalized reasoning” remains a hypothesis supported by the performance gain but contradicted by the visual quality of the intermediate steps.

Finally, the abstract implies a comprehensive superiority over existing methods. However, Table 1 (Section 4) reveals that on the “Real+Arr” split, the Gemini 3 Flash model achieves 96.2% accuracy, significantly outperforming the proposed Bagel+Mixed model (66.5%). The claim of outperforming “all baselines” is therefore factually incorrect based on the paper’s own data. The text must be revised to accurately reflect that the method outperforms baselines on synthetic and specific real-world splits, but not universally across all tested conditions.

Action items.

- **[writing]** Title/Abstract claim general ‘spatial reasoning’ enhancement, but results (Table 1) are limited to AI2-THOR, Habitat, VST, and Matterport3D. Narrow scope to tested environments or add cross-domain evaluation.
- **[writing]** Conclusion claims ‘internalized spatial reasoning,’ yet Fig 3 admits generated thoughts are ‘spatially imprecise’ and ‘lack structure.’ Hedge claim to ‘improves performance’ rather than asserting a verified cognitive mechanism.
- **[writing]** Abstract claims superiority over ‘all baselines,’ but Table 1 shows Gemini 3 Flash (96.2%) beats Bagel+Mixed (66.5%) on ‘Real+Arr.’ Correct to ‘outperforms most baselines’

on synthetic splits’ or acknowledge the gap.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a method for enhancing spatial reasoning in multimodal models using “Imaginative Perception Tokens” (IPT) to generate intermediate visual representations (e.g., imagined side-views or top-down maps) during inference. The research relies on synthetic data generated from established simulation environments (AI2-THOR, Habitat, ProcTHOR) and public, pre-existing real-world datasets (Matterport3D, Visual Spatial Tuning, ScanNet++, MessyTable).

From a safety and ethics perspective, the work is low-risk. The methodology does not involve:

1. **Human Subjects:** No new human-subjects data collection, surveys, or behavioral experiments are described. The “human review” mentioned in Section `supp_data_pt` refers to quality control of synthetic annotations, not interaction with human participants requiring IRB oversight.
2. **PII or Sensitive Data:** The datasets used are standard benchmarks in computer vision and robotics, which are publicly available and do not contain personally identifiable information (PII) or sensitive medical/financial records.
3. **Dual-Use Harm:** The capability to “imagine” spatial layouts is a reasoning enhancement for navigation and scene understanding. It does not lower the barrier to generating disinformation, biological/chemical hazards, or cyber-attacks. The generated images are abstract spatial representations (often low-fidelity or grayscale in early experiments) intended for internal model reasoning, not for creating deceptive deepfakes or surveillance tools.
4. **Data Licensing:** The paper cites standard licenses for the public datasets used (e.g., Matterport3D, ScanNet++) and does not appear to scrape data in violation of Terms of Service.

There are no missing disclosures regarding consent, conflicts of interest, or responsible release of harmful capabilities. The paper does not present operational details for exploits or vulnerabilities. Consequently, no action items are required.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method for spatial reasoning, but the experimental design currently fails to rule out several alternative explanations for the reported gains.

First, the primary evidence for the efficacy of Imaginative Perception Tokens (IPTs) relies on single-run results reported in Table 1 and the supplementary Table `pt_breakdown`. The authors report specific accuracy improvements (e.g., Bagel + IPT achieving 61.1% vs. Bagel base at 36.3% on EgoDir) without providing standard deviations, confidence intervals, or results across multiple random seeds. In spatial reasoning tasks, performance can fluctuate significantly based on initialization and sampling. Without variance reporting, it is impossible to determine if these gains are robust or artifacts of a lucky seed. The authors must re-run experiments across at least 3-5 seeds to establish statistical stability.

Second, the ablation study in Table `pt_breakdown` contains a critical confound regarding the source of improvement. The “Mixed Training” variant (which includes 15k real-world VST examples) achieves 71.7% on EgoDir, significantly outperforming the “+ IPT” variant (61.1%) which uses only synthetic data. The paper attributes the success to the IPT mechanism, yet the data suggests that simply increasing the volume and diversity of training data (the “Mixed” component) yields a larger boost than the IPT tokens themselves. The design does not isolate the contribution of the token architecture from the contribution of the expanded dataset. A fair comparison requires a control run where the model is trained on the full “Mixed” dataset using standard answer-only tokens (no IPTs) to see if the data alone explains the performance jump.

Finally, the comparison against frontier models (GPT-5, Gemini 3) is unfair due to training asymmetry. The proposed method is heavily fine-tuned on a custom dataset, while the baselines are evaluated in zero-shot or few-shot modes. This conflates the benefit of the proposed method with the benefit of task-specific fine-tuning. To support the claim that IPTs are a superior *method* for spatial reasoning, the authors must compare their approach against a baseline that receives the same fine-tuning regimen (same data, same compute) but lacks the IPT mechanism. Without these controls, the evidence supports “fine-tuning on this specific dataset helps” more than “IPTs enhance spatial reasoning.”

Action items.

- **[science]** The paper presents a compelling method for spatial reasoning, but the experimental design currently fails to rule out several alternative explanations for the reported gains. First, the primary evidence for the efficacy of Imaginative Perception Tokens (IPTs) relies on single-run results reported in Table 1 and the supplementary Table `pt_breakdown`. The authors report specific accuracy improvements (e.g., Bagel + IPT achieving 61.1% vs. Bagel base at 36.3% on EgoDir) without providing standard de

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the results in this paper is currently insufficient to support the quantitative claims made. While the experimental design (data curation, benchmarks) is detailed, the analysis of the resulting performance metrics lacks necessary rigor regarding uncertainty and inferential testing.

1. Missing Uncertainty Reporting (Primary Concern): Throughout the results sections, specifically in Table 1 (main paper) and Table~\ref{tab:pt_breakdown} (supplementary), the authors report single-point accuracy percentages (e.g., “73.5%”, “61.1%”) for their proposed BAGEL models. There is no accompanying standard deviation (SD), standard error (SE), or confidence interval (CI), nor is there a mention of the number of random seeds used to generate these numbers. In the context of training deep learning models, performance can vary significantly based on random initialization, data shuffling, and optimization noise. Reporting a single number implies a level of precision and stability that is not statistically justified. Without variance estimates, a reader cannot determine if the reported “improvement” over baselines is a robust effect or a lucky run. This is a critical gap for any claim of “enhanced” performance.

2. Unsubstantiated Claims of Significance: The paper frequently uses language implying statistical superiority (e.g., “outperforms,” “improves”) based solely on the magnitude of the point estimate difference. For instance, in Table~\ref{tab:pt_breakdown}, the “Bagel (label-only)” model shows a 37.2 percentage point jump over “Bagel (base)” on the EgoDir split. While this is a large difference, the paper does not perform or report a hypothesis test (such as a paired t-test or a bootstrap test) to confirm that this difference is statistically significant. Furthermore, with multiple splits (EgoDir, Path, PathArr, Real, Real+Arr) and multiple model variants being compared, the risk of Type I errors (false positives) is high. The authors should either run appropriate statistical tests with multiple-comparison corrections (e.g., Holm-Bonferroni) or soften their language to describe “observed improvements” rather than definitive “significant” gains.

3. Inconsistent Reporting in Ablation Studies: In the supplementary experiments with Qwen2.5-VL (Tables~\ref{tab:qwen_discrete} and~\ref{tab:qwen_modality}), the authors report accuracy changes (e.g., 55.0% to 59.6% with grayscale) but again fail to report the number of seeds or variance. Since these are controlled ablation studies where the only variable is the token representation, the lack of variance reporting is particularly problematic. It prevents the community from assessing the reliability of the finding that “grayscale improves PT.”

Recommendation: The authors must re-run their primary experiments (BAGEL models) with at least 3 independent random seeds and report the mean $\pm$ standard deviation for all accuracy metrics. Additionally, they should perform statistical significance testing on the key comparisons and report the p-values (with correction for multiple comparisons) or explicitly state that no formal significance testing was performed and that the results are descriptive. Without these additions, the quantitative claims of the paper remain statistically unsupported.

Action items.

- **[science]** Table 1 (main) and Table~\ref{tab:pt_breakdown} report single-point accuracy percentages (e.g., 73.5%, 61.1%) for the proposed BAGEL models without any measure

of variance (SD, SE, or CI) or number of seeds. In deep learning, single-run reporting is insufficient to distinguish signal from stochastic noise. Report mean $\pm$ SD over at least 3 independent training seeds for all primary results, or explicitly state that results are from a single run and treat them as preliminary.

- **[science]** The paper claims ‘significant’ improvements (e.g., Bagel label-only vs. base in Table~\ref{tab:pt_breakdown}) based on point estimates alone. No hypothesis tests (e.g., paired t-tests or bootstrap tests) are reported to validate these differences. Given the multiple comparisons across 5 splits and 3 model variants, apply a statistical test with appropriate multiple-comparison correction (e.g., Holm-Bonferroni) or rephrase claims to ‘observed improvement’ without invoking statistical significance.
- **[writing]** Table~\ref{tab:qwen_discrete} and Table~\ref{tab:qwen_modality} report accuracy differences (e.g., 55.0% vs 59.6%) for Qwen2.5-VL experiments without reporting the number of seeds or variance. Since these are ablation studies on model architecture (codebook size, grayscale), the lack of uncertainty reporting makes it impossible to determine if the observed gains are robust or due to random initialization variance. Report variance across seeds for these specific ablation results.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript generally presents a clear narrative flow, with well-structured sections that guide the reader through the methodology, data curation, and results. The use of specific subsections for different data sources (AI2-THOR, Habitat, etc.) aids in digesting the complex data generation pipeline. However, several areas require attention to ensure the prose is frictionless and the structural elements are complete.

First, the section “Imaginative Token Exploration with Different VLMs” (Section \ref{supp_vlms}) begins abruptly with technical details about VQ-GAN configurations. A strong topic sentence is missing that would frame the section’s purpose: to evaluate discrete tokens before pivoting to the continuous latent approach used in the main model. Adding a sentence like “We first explored discrete imaginative perception tokens using VQ-GANs on Qwen2.5-VL to assess feasibility...” would immediately orient the reader.

Second, within the “Data Curation Details” (Section \ref{supp_data}), the “Path Tracing” subsection launches directly into “AI2-THOR” without a brief introductory sentence summarizing the overall data strategy (e.g., combining synthetic and real-world sources). A lead-in sentence would help the reader understand the scope before diving into specific datasets.

Third, there are minor structural inconsistencies in the provided text chunks regarding table captions and completeness. For instance, the caption for Table 1 (Training modes) and the

table structure itself appear fragmented in the first chunk, and the caption for the Path Tracing results table (Table \ref{tab:pt_breakdown}) is split or missing in parts of the provided text. While this may be an artifact of the chunking process, the final manuscript must ensure all table captions are complete, correctly placed, and fully visible to avoid reader confusion.

Finally, the “Text CoT generation” paragraph in the first chunk is a sentence fragment (“Targets generated by GPT-5.1...”). This should be rewritten as a complete sentence or integrated into a bulleted list to maintain the professional tone of the surrounding text. Addressing these points will ensure the paper reads smoothly and that all structural elements are polished.

Action items.

- **[writing]** The manuscript generally presents a clear narrative flow, with well-structured sections that guide the reader through the methodology, data curation, and results. The use of specific subsections for different data sources (AI2-THOR, Habitat, etc.) aids in digesting the complex data generation pipeline. However, several areas require attention to ensure the prose is frictionless and the structural elements are complete. First, the section “Imaginative Token Exploration with Different VLMs” (Secti